

Chapter 5: Mathematics Logic

Class 23 – Revision, Mathematics Logic Assessment & Problem-Solving Evaluation

Learning Objectives

After completing this lesson, students will be able to:

- Revise all concepts learned in **Mathematics Logic**.
 - Apply logical reasoning to solve mathematical problems.
 - Demonstrate mental mathematics skills with speed and accuracy.
 - Solve number patterns, visual puzzles, and brain teasers confidently.
 - Participate in mathematics quizzes, team activities, and oral assessments.
 - Improve critical thinking, observation, and problem-solving abilities.
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Introduction

This lesson is a **revision and assessment session** designed to review all the concepts covered in **Chapter 5: Mathematics Logic**. Students will strengthen their understanding through quizzes, puzzles, mental mathematics, logical reasoning activities, and practical problem-solving tasks.

Revision helps students improve confidence, accuracy, and mathematical thinking while preparing them for the final assessment.

Revision of Number Patterns

Students revise:

- Number Patterns
- Number Sequences
- Missing Numbers
- Even and Odd Numbers
- Skip Counting
- Ascending Order
- Descending Order
- Shape Patterns

Examples

- 2, 4, 6, 8, __
 - 5, 10, 15, __
 - 100, 90, 80, __
 - 8, 16, 24, __
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Revision of Logical Reasoning

Students revise:

- Logical Thinking
- Visual Reasoning
- Pattern Recognition
- Shape Matching
- Odd One Out
- Mirror Images
- Direction Sense
- Analogies

Example

Teacher : School :: Doctor : _____

Answer: Hospital

Revision of Visual Puzzles

Students practice:

- Picture Matching
 - Shape Sequences
 - Complete the Pattern
 - Find the Difference
 - Missing Shape Activities
 - Picture Observation
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Revision of Mental Mathematics

Students revise:

- Mental Addition
- Mental Subtraction
- Multiplication
- Division
- Estimation
- Number Comparison
- Time Calculations

Examples

- $45 + 35 = 80$
 - $90 - 25 = 65$
 - $9 \times 8 = 72$
 - $72 \div 8 = 9$
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Revision of Strategy Games

Students revise:

- Sudoku (Basic)
 - Magic Squares
 - Number Bingo
 - Puzzle Race
 - Number Relay
 - Tangram Puzzle
 - Memory Card Game
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Revision of Problem-Solving Skills

Students solve real-life mathematical problems.

Examples

Question 1

Ravi has **24 pencils**. He buys **16 more**.

How many pencils does he have now?

Answer: 40 pencils

Question 2

A shopkeeper has **80 chocolates**. He sells **35 chocolates**.

How many chocolates are left?

Answer: 45 chocolates

Question 3

A bus has **50 seats**. **42 students** are seated.

How many seats are empty?

Answer: 8 seats

Mathematics Logic Quiz

Students answer questions from all topics.

Sample Quiz Questions

1. What is a number pattern?
2. Name three even numbers.
3. What comes next: 10, 20, 30, ____?
4. Which number is greater: 86 or 68?
5. What is the opposite direction of North?
6. Complete the analogy: Bird : Nest :: Bee : _____.
7. What is 12×5 ?
8. What is $100 \div 10$?
9. Why is logical reasoning important?
10. Name one strategy game.

Brain Teaser Practice

Students solve fun thinking puzzles.

Examples

1. Which number comes before **200**?

Answer: 199

2. Which is heavier: **1 kg of cotton** or **1 kg of iron**?

Answer: Both weigh the same.

3. I have keys but cannot open doors. What am I?

Answer: A Keyboard

4. What comes once in a minute, twice in a moment, but never in an hour?

Answer: The letter M

Team Logic Challenge

Students work in groups to solve mathematical challenges.

Activities

- Number Pattern Race
 - Sudoku Challenge
 - Picture Puzzle Competition
 - Shape Matching
 - Mental Maths Relay
 - Brain Teaser Round
 - Problem-Solving Challenge
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Oral Reasoning Assessment

Students answer oral questions individually.

Sample Questions

- What is a number sequence?
 - Name five odd numbers.
 - What is estimation?
 - Why is logical thinking important?
 - How do puzzles improve our brain?
 - What is skip counting?
 - Give one example of a real-life math problem.
 - What is Sudoku?
 - Name one mathematical strategy game.
 - Why should we check our answers?
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Practical Assessment

Students demonstrate the skills learned throughout the chapter.

Practical Tasks

Students should be able to:

- Complete number patterns.
 - Solve missing number activities.
 - Identify odd and even numbers.
 - Arrange numbers in ascending and descending order.
 - Solve visual puzzles.
 - Complete a Sudoku puzzle.
 - Solve mental mathematics problems.
 - Explain logical reasoning answers.
 - Participate in team activities confidently.
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Teacher Feedback & Improvement Tips

The teacher evaluates:

- Number Pattern Skills
- Mental Mathematics
- Logical Thinking
- Problem-Solving Ability
- Observation Skills
- Accuracy
- Speed
- Participation

Improvement Tips

- Practice mental calculations every day.
 - Solve puzzles regularly.
 - Observe patterns carefully.
 - Read questions completely before answering.
 - Check answers before submitting.
 - Participate actively in classroom activities.
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Did You Know?

- Solving puzzles every day improves memory and concentration.
 - Mental mathematics makes everyday calculations faster.
 - Logical reasoning is used in science, engineering, computer programming, and many other careers.
 - Regular practice strengthens critical thinking and decision-making skills.
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Points to Remember

- Look for patterns before solving.
- Think carefully and logically.
- Use mental mathematics to calculate quickly.
- Read questions carefully.
- Check your answers for accuracy.
- Practice regularly to improve speed and confidence.

